

Adaptive Texture-Aware Pixel Selection for Robust Direct Visual Odometry in UAV Navigation

Jeremias Gaia¹, Leonardo Fagundes-Junior², Carlos Soria¹, and Alexandre Brandão²

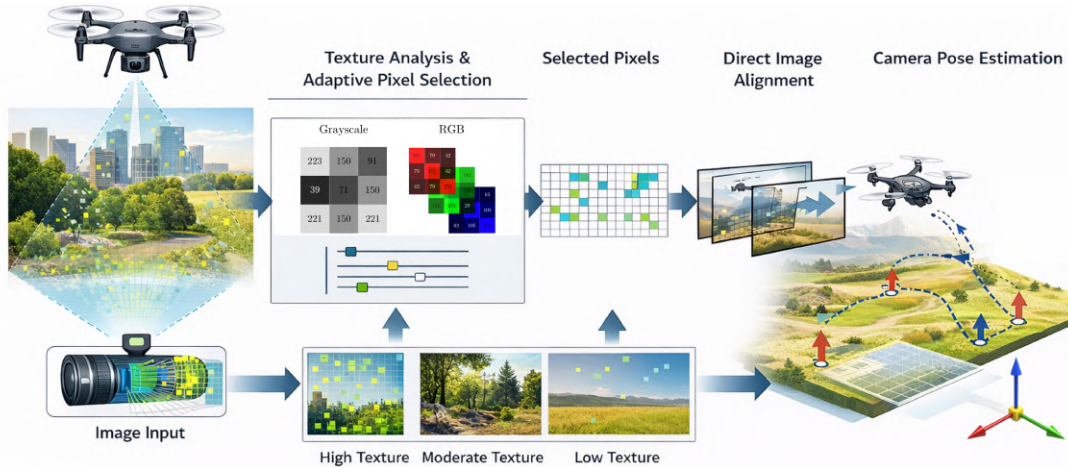


Fig. 1: Illustration of the proposed visual odometry in scenarios with different texture levels.

Abstract—Direct visual odometry (DVO) is becoming widely used to estimate camera ego-motion and simultaneously reconstruct/map the environment from image sequences, particularly due to its efficient use of photometric information and strong performance in weak-texture scenarios. However, since DVO relies directly on pixel intensity patterns for frame-to-frame alignment, it is inherently sensitive to brightness inconsistencies and illumination changes, which can degrade pose estimation, particularly in aerial robotics scenarios involving motion blur, illumination variations, and texture-poor environments. This paper proposes an adaptive pixel selection mechanism for DVO systems. The proposed approach prioritizes regions with strong structural content, ensuring that the pose estimate remains constrained by meaningful image gradients, even in low-texture or highly homogeneous scenarios. By integrating image texture descriptors, the system dynamically adjusts selection parameters, according to the scene complexity. This enables the operation across structured, semi-structured, and unstructured environments without introducing significant computational overhead. Experimental validation in both indoor and outdoor scenarios demonstrates that the proposed method preserves uniform spatial coverage while suppressing low-gradient and photometrically unstable regions. The results demonstrate enhanced trajectory consistency and greater robustness under challenging visual conditions, confirming the algorithm effectiveness for UAV navigation in GPS-denied environments.

Index Terms—Mobile Vehicle Visual Odometry, Pose Estimation, Texture Analysis.

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¹Ing. Gaia and Dr. Soria are with the Instituto de Automática, at Facultad de Ingeniería, Universidad Nacional de San Juan (UNSJ-CONICET), Av. Libertador Gral San Martín 1109 (O), San Juan - SJ, Argentina. ²M.Sc. Fagundes-Junior and Dr. Brandão are with the Núcleo de Especialização em Robótica (NERo), Department of Electrical Engineering and Department of Informatics, and Graduate Program on Computer Science, Universidade Federal de Viçosa, Viçosa - MG, Brazil. E-mails: {jgaia, soria}@inaut.unsj.edu.ar and {leonardo.fagundes, alexandre.brandao}@ufv.br.

I. INTRODUCTION

Autonomous aerial robotics has experienced significant growth in recent years, driven by applications such as infrastructure inspection [1], search and rescue [2], environmental monitoring [3], and warehouse automation [4]. In many of these scenarios, unmanned aerial vehicles (UAVs) must operate in GPS-denied or degraded environments, requiring on-board perception systems capable of reliable state estimation. Simultaneous Localization and Mapping (SLAM) provides a principle-based framework for jointly estimating vehicle pose and reconstructing the surrounding environment [5], [6]. These systems typically rely on the fusion of data from various sensors, such as monocular and stereo cameras, LiDAR, and inertial measurement units (IMUs) [7], [8].

Among the various SLAM modalities, visual SLAM (V-SLAM) is particularly attractive for aerial platforms due to the favorable weight, cost, and energy characteristics of cameras compared to LiDAR systems. When relying exclusively on camera input, pose estimation is performed through visual odometry (VO), which refers to the process of estimating an agent's egomotion (robot, human, vehicle, etc.) using input from one or more cameras [9], [10]. Visual methods are typically categorized into indirect (feature-based) and direct approaches. Indirect methods rely on sparse feature extraction and matching, estimate motion by using pixel intensities and positions, whereas direct methods estimate motion by minimizing photometric error over selected pixel intensities, first construct a scene representation prior to motion estimation.

Direct visual odometry methods such as LSD-SLAM [11], Stereo LSD-SLAM [12], and Direct Sparse Odometry

(DSO) [13] have demonstrated strong performance in texture-rich environments taking advantage of photometric consistency and semi-dense or sparse pixel selection based on gradient. These approaches directly optimize camera pose by aligning pixel intensities across frames, avoiding explicit feature extraction and descriptor matching.

However, UAV deployment introduces additional challenges: *i)* high-speed motion causing motion blur; *ii)* large viewpoint changes; *iii)* exposure variations due to illumination dynamics *iv)* and frequent operation in low-texture environments (e.g., corridors, walls, or agricultural fields). Under such conditions, conventional gradient-based pixel selection strategies may either oversample high-gradient clusters or include poorly conditioned pixels that degrade photometric optimization. Since evaluating all image pixels is computationally infeasible in real-time systems, pixel selection becomes a critical component of direct methods [14]–[17].

In contrast to prior approaches that rely solely on gradient magnitude or learned priors, this work address this limitation by proposing a lightweight adaptive pixel selection strategy designed to improve structural observability in aerial direct visual odometry, while explicitly measures local texture richness and dynamically adjusts pixel selection parameters. Our method identifies pixels that are well-suited for recovering a robot's trajectory, with potential extensions to direct odometry-based SLAM systems. Our strategy improves structural conditioning of the photometric optimization while preserving real-time feasibility for UAV deployment.

The main contributions of this paper are: *i)* an adaptive texture-aware pixel selection mechanism for direct visual odometry; *ii)* a strategy to preserve spatial uniformity while prioritizing structurally informative regions; *iii)* experimental validation in indoor and outdoor scenarios demonstrating improved robustness in low-texture and high homogeneous environments.

The remainder of this article is organized as follows: Section II introduces the theoretical concepts used to develop the proposed algorithm, which is described in detail in Section III. Section IV presents the results of field experiments, and Section V provides concluding remarks and directions for future works.

II. MATERIALS AND METHODS

A. Theoretical Background

This section outlines the statistical computations applied to input images, which support the design of the proposed system.

1) *Gray Level Co-occurrence Matrix (GLCM)*: In the early 1970s, Robert Haralick introduced the Gray Level Co-occurrence Matrix (GLCM) as a method for extracting textural features from images [18]. According to this method, the spatial relationship between gray levels in an image I encodes its texture information. The GLCM quantifies the frequency of gray-level transitions in a specified direction. Typically, it is computed across multiple orientations (e.g., 0° , 45° , 90° , and 135°) to capture texture features in different directions.

In this work, we focus on one of the fourteen statistical measures proposed by Haralick: image *Homogeneity*, also known as the inverse difference moment. As the name suggests, this metric provides an estimate of the uniformity of gray levels, offering insight into the amount of texture in an image. It is defined as:

$$H = \sum_{i=1}^{N_g} \sum_{j=1}^{N_g} \frac{1}{1 + (i - j)^2} p(i, j) \quad (1)$$

where $p(i, j)$ denotes the $(i, j)^{\text{th}}$ element of the normalized gray-level co-occurrence matrix P , and N_g is the number of gray levels in the image.

A high homogeneity value indicates that the image contains large regions with similar gray-level intensities, implying low texture content.

2) *Gray Level Run Length Matrix*: One year after Haralick's study, Galloway introduced the Gray Level Run Length Matrix (GLRLM) to analyze image texture [19]. Unlike the Gray Level Co-occurrence Matrix (GLCM), this method quantifies the extent of consecutive runs of pixels that share the same gray level along a specific direction in the image. The resulting matrix, denoted as $v(i, j)$, records the frequency of runs of length j for pixels with gray level i . Similar to the GLCM, the GLRLM is computed across multiple orientations. From this matrix, various statistical features can be derived to describe the image texture, including *short run emphasis* (SRE) and *long run emphasis* (LRE).

Given a GLRLM V , the element $v(i, j)$ represents the number of occurrences of a run of length j at gray level i . Let N_g be the number of gray levels, and N_r the number of run lengths considered. The total number of runs in the image, N_{runs} , is defined as:

$$N_{runs} = \sum_{i=1}^{N_g} \sum_{j=1}^{N_r} v(i, j) \quad (2)$$

Based on this, the short run emphasis (SRE) and long run emphasis (LRE) are computed as follows:

$$SRE = \frac{\sum_{i=1}^{N_g} \sum_{j=1}^{N_r} \frac{v(i, j)}{j^2}}{N_{runs}} \quad (3)$$

$$LRE = \frac{\sum_{i=1}^{N_g} \sum_{j=1}^{N_r} v(i, j) \cdot j^2}{N_{runs}} \quad (4)$$

The Short Run Emphasis (SRE) metric, defined in Eq. (3), assigns an inverse quadratic weight to each run length via the term $1/j^2$. As a result, shorter runs exert a stronger influence on the final SRE value. This behavior allows SRE to capture fine-grained variations in image texture. Specifically:

- High SRE values correspond to a predominance of short gray-level runs, which are indicative of fine textures with frequent intensity transitions.
- Low SRE values reflect a higher occurrence of longer runs, suggesting a coarser texture with less frequent intensity changes.

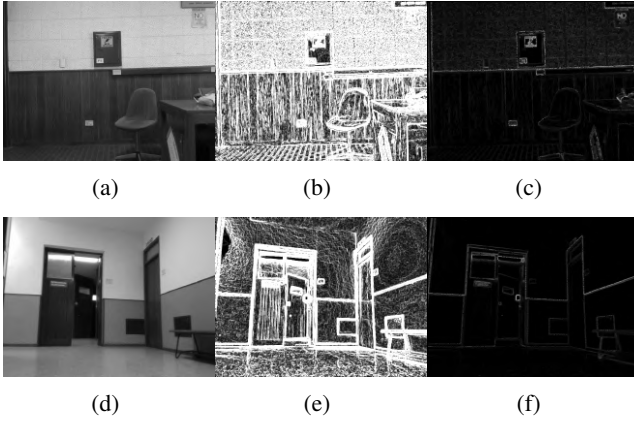


Fig. 3: Comparison between the gradients extracted using different techniques. (a)-(d): Grayscale versions of the input images. (b)-(e): Gradient images obtained using the Sobel operator. (c)-(f): G_{max} images.

In contrast, the Long Run Emphasis (LRE) metric is designed to measure the presence of extended uniform regions within an image. It applies a direct quadratic weight to each run length using the term j^2 , thereby assigning greater importance to longer runs:

- High LRE values signify a predominance of long gray-level runs, typical of coarse textures with large homogeneous areas or gradual transitions in intensity.
- Low LRE values indicate a larger proportion of short runs, which is characteristic of finer textures.

Together, SRE and LRE serve as complementary descriptors that quantify different aspects of image texture. While SRE emphasizes fine textural details and rapid local variations, LRE highlights the presence of coarse structures and broader homogeneity. The combined analysis of both metrics offers a robust and comprehensive characterization of the textural properties of an image.

B. Maximum Gradients Image

Edges in an image correspond to locations where abrupt intensity transitions occur, typically indicating structural boundaries of objects within the scene. These edges are commonly detected by computing spatial derivatives, which highlight changes in gray levels. However, since differentiation inherently amplifies high-frequency components, it can also enhance image noise, leading to spurious or fragmented edges. Therefore, applying noise suppression techniques prior to edge detection is essential to preserve the integrity of meaningful features.

In the context of pose estimation, edges play a critical role as they denote regions of high structural significance. Accurately identifying such features enables robust tracking of camera motion. To this end, we employ a filtering approach known as the *maximum gradients image*, designed to emphasize prominent edges while suppressing weak and noisy responses. The process begins with the computation of the gradient magnitude G from the input image. This is

defined as:

$$G = \sqrt{G_x^2 + G_y^2},$$

where G_x and G_y are the gradients in the horizontal and vertical directions, respectively. The resulting magnitude image is normalized to the range $[0, 255]$, and a threshold Th_f is applied to retain only regions exhibiting sufficiently high gradient intensity. This step effectively filters low-contrast areas that do not contribute meaningful structural information.

To further refine the edge representation, we apply non-maximum suppression in two stages: first along the vertical direction and then along the horizontal. This technique ensures that only local maxima in the gradient field are preserved, enhancing the sharpness and spatial accuracy of the detected edges. Subsequently, morphological erosion is used to eliminate residual noise and to thin the edge contours, producing a binary image that highlights only the most salient edge points.

This entire process is independently applied to each channel of the input RGB image. Once the maximum gradient response is computed for each channel, the results are combined to generate the final maximum gradient image G_{max} . This fusion ensures that edges prominent in any of the color channels are captured in the final output.

Figures 3(c) and (g) show the limitations of using raw image derivatives without preprocessing, where minor intensity fluctuations can be incorrectly interpreted as valid edges. In contrast, Figures 3(d) and (h) demonstrate the effectiveness of the proposed maximum gradient extraction method in producing a cleaner and more structurally meaningful edge map.

III. ADAPTIVE KEYPOINT SELECTION ALGORITHM

An overview of the keypoint extraction algorithm for pose estimation is presented in Figure 4. The algorithm takes as input the RGB image and its grayscale counterpart. The RGB version is used to compute G_{max} , while the grayscale image undergoes Gaussian blur filtering and Canny edge detection to produce a binary edge image.

A *Patch Extractor* module divides each image into patches of size S_{ROI} (initially set to $S_{patch} = 16$), which are subsequently evaluated by the *Patch Selector*. At the end of the process, the number of extracted keypoints is evaluated. If this number, denoted N_{points} , does not meet a predefined minimum, the algorithm automatically adjusts either S_{ROI} or Th_f accordingly.

A. Adaptive Threshold Adjustment (Th_f)

The threshold variable Th_f directly influences the number of high-gradient points in G_{max} that are eligible for selection as keypoints. High values of Th_f reduce the appearance of gradients in G_{max} , while low values allow low-gradient regions to emerge. An excessively high threshold can lead to significant information loss, reducing the number of potential keypoints. Conversely, an overly low threshold may result in the selection of low-gradient points, which negatively impact pose estimation. These trade-offs highlight the need for an adaptive thresholding mechanism.

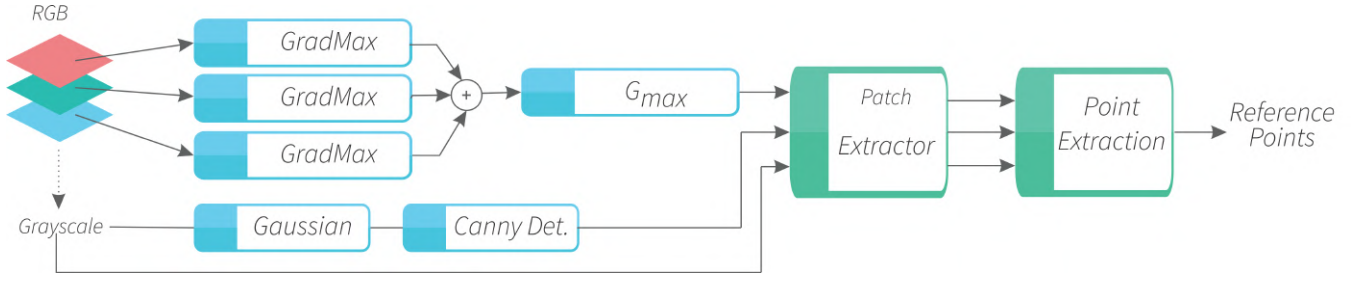


Fig. 4: Overview of the system.

At system initialization, Th_f is set to a default value, an initial keypoint extraction is performed using a fixed window size S_{ROI} , and the resulting number of points is evaluated. If N_{points} is insufficient, the threshold is decreased. The reduction percentage r_{perc} is computed based on the homogeneity measures defined in Equation (1), as follows:

$$r_{perc} = \frac{1}{rows \cdot cols} \sum_{i=1}^{rows} \sum_{j=1}^{cols} g(i, j) + \left(\frac{H^{0^\circ} + H^{90^\circ}}{2} \right) \quad (5)$$

where $g(i, j)$ denotes the gradient magnitude at pixel (i, j) in G_{max} . High homogeneity values indicate that the threshold applied to generate G_{max} has been too restrictive and must be relaxed. Thus, r_{perc} increases proportionally with the average homogeneity in the horizontal (H^{0°) and vertical (H^{90°) directions. The threshold is then updated as:

$$Th_{new} = Th_f \cdot r_{perc} \quad (6)$$

A lower bound is defined for Th_f , below which the threshold is no longer reduced; instead, the search window size S_{ROI} is modified.

B. Adaptive Search Window Adjustment (S_{ROI})

The value of S_{ROI} determines the size of the search window for keypoint extraction. A fixed window size may lead to several issues:

- An excessively large window may result in too few extracted keypoints, failing to meet the minimum requirement.
- A window that is too small increases computational cost and may cause significant delays.
- In regions rich in texture or structure, a large window may underutilize the information, yielding few keypoints.
- Conversely, very small windows may lead to excessive keypoint extraction, slowing down subsequent pose estimation steps.

To address these issues, the window size is adaptively adjusted based on image characteristics. As illustrated in the *Patch Extractor* block of Figure 4, the algorithm processes the G_{max} , Canny edges, and grayscale versions of the input image, dividing them into patches of size S_{ROI} . Each patch is further subdivided into four sub-patches of size $S_{patch}/2$ to ensure spatial distribution of keypoints. The system is configured to extract a maximum of two keypoints per patch.

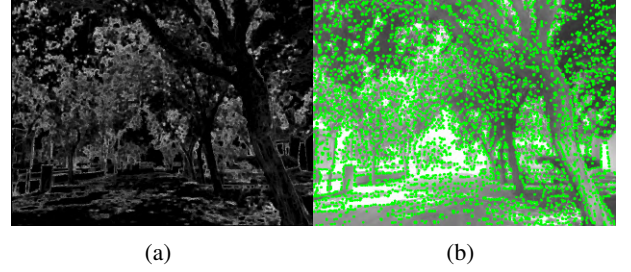


Fig. 5: Extreme case in which the system extracts keypoints uniformly across the image after exhausting all adjustments to Th_f and S_{ROI} , and increasing to three keypoints per patch. (a): G_{max} image. (b): Extracted keypoints.

If the desired number of keypoints is not obtained and Th_f has already reached its lower bound, S_{ROI} is halved. If this is still insufficient, the system attempts to extract three keypoints per patch to increase the likelihood of reaching the target.

In extreme cases, illustrated in Figure 5, where neither reducing the window size nor lowering Th_f suffices, the system resorts to extracting keypoints in a spatially uniform manner across the entire image. This ensures sufficient spatial coverage to capture camera motion.

C. Keypoint Extraction

The keypoint extraction module receives corresponding patches from the G_{max} , the Canny edges, and the grayscale images. Each patch is individually evaluated to determine whether it meets the criteria for reliable keypoint selection.

First, the grayscale patch is assessed using the SRE metric defined in Equation (3). If $SRE \geq SRE_{max}$, the patch is discarded, as this indicates the presence of excessive gray-level variations, often corresponding to small or distant textures, which are unreliable for pose estimation.

Next, the binary patch derived from the Canny edge detector is analyzed using the LRE metric from Equation (4). The Canny image is used due to its binarized nature, which facilitates efficient computation of long edge sequences. If $LRE \leq LRE_{min}$, the patch is also discarded.

Only patches satisfying that $SRE < SRE_{max}$ and $LRE_{min} < LRE$ are deemed suitable for keypoint extraction. These conditions ensure the presence of nearby objects or structured elements that contribute to accurate pose estimation.

The reference thresholds of $SRE_{max} = 0.9$ and $LRE_{min} = 30$ were empirically determined through experimentation, representing a trade-off between performance and computational efficiency that yields robust results in both indoor and outdoor environments. The Canny edge detector also needs a threshold ($canny_thr$) to filter out weak edges. This value was set to 100. Section IV-B shows an ablation study to assess the impact of these parameters on the performance of the system.

IV. EXPERIMENTS AND RESULTS

This section presents qualitative and quantitative results obtained from evaluating our method in real-world field experiments using proprietary datasets, as well as on benchmark datasets.

To evaluate the proposed method under diverse conditions, three datasets were collected during field experiments on the campus of the Facultad de Ingeniería at the Universidad Nacional de San Juan, covering indoor, outdoor, and mixed scenarios. As these datasets lack ground-truth measurements, they are used exclusively for qualitative evaluation. Quantitative evaluation and comparison with a state-of-the-art method are conducted using the TUM RGB-D benchmark dataset.

An operating range was defined for the number of points generated by the selection system and used for pose estimation. In the experiments, the number of points was constrained to lie between 2000 and 3600. The lower bound reflects the minimum required for proper operation of the pose estimation system, while the upper bound enforces a relaxation of the adaptive threshold.

A. Qualitative Results

Indoor Experiments were conducted in an office environment, characterized by hallways with extensive low-gradient and homogeneous areas. Figures 6 and 7 illustrate dataset frames corresponding to two representative scenarios.

Figure 6 illustrates an instance from the dataset in which, as the system moves along the corridor, it transitions from extracting two points per patch to extracting three. This adaptive behavior is visually reflected in a higher concentration of selected points around structured regions (Figures 6(c), 6(f)), effectively avoiding low-gradient areas.

Figure 7 illustrates a scenario where the camera moves toward a wall. As the distance decreases, homogeneous regions increasingly dominate the frame; however, the system concentrates point extraction around structural elements, including window frames and baseboards.

Outdoor environments pose several challenges for visual systems, including varying lighting conditions, shadowed areas, foliage, streets, and dynamic objects.

Figure 8 demonstrates how the pixel selection strategy performs under such conditions. Even in high-gradient regions, such as the contrast between bright sky and dark foliage, the system reduces the number of extracted points and prioritizes structured elements such as window frames, tree trunks, and lampposts.

This behavior results from the *Point Extraction* module. Regions dominated by rapid gray-level transitions (i.e., high SRE and low LRE), such as foliage, are largely avoided by the algorithm. Nevertheless, due to the trade-off between performance and computational efficiency, the system may still select points in suboptimal regions during outdoor scenarios. Importantly, it consistently maintains a spatially uniform point distribution and largely avoids low-gradient homogeneous areas (e.g., sky, walls, grass).

B. Ablation study

To thoroughly evaluate the individual contribution of each component within the proposed adaptive pixel selection mechanism, a comprehensive ablation study was conducted. The primary objective of this analysis is to quantify how

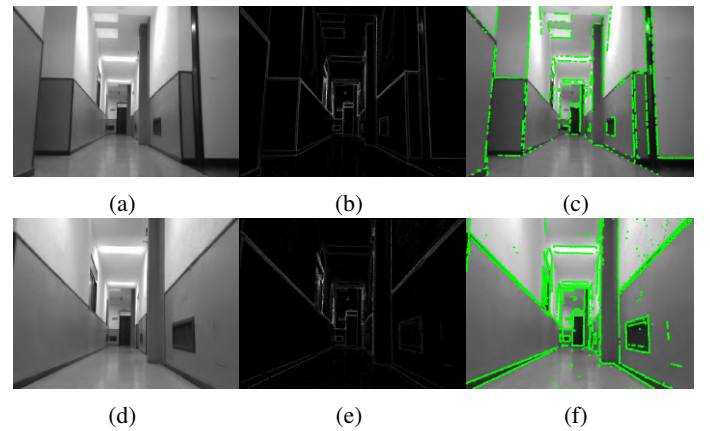


Fig. 6: Instance in which the system must extract three keypoints per patch after minimizing both Th_f and S_{ROI} . (a)-(d): Grayscale input images. (b)-(e): G_{max} images. (c)-(f): Extracted keypoints.

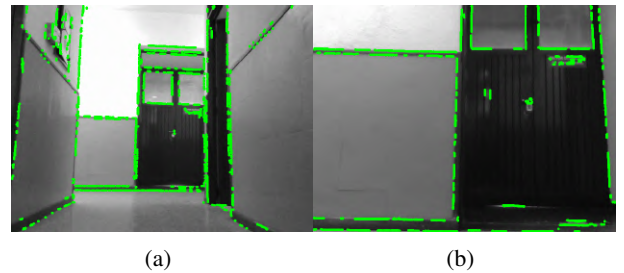


Fig. 7: Close-up of a wall in an office environment. As homogeneous regions increase, the algorithm intensifies point extraction along edges.

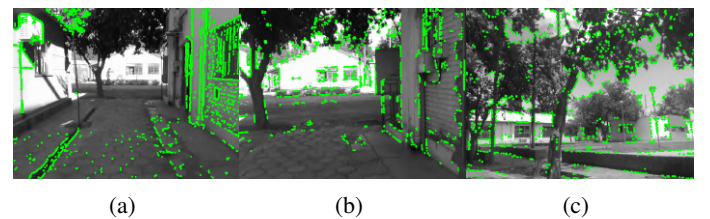


Fig. 8: Results for outdoor tests.

TABLE I: Ablation Study Results.
The selected system parameters appear in boldface.

Parameter	Value	N° Extracted Points (Mean)	Frames outside Op.Band [%]	Point Retrieval Time [s]	Pose Estimation Time [s]	ATE RMSE [m]
<i>canny_thr</i>	25	2803	10.00	0.0876	0.0828	0.2137
<i>canny_thr</i>	50	2858	11.52	0.0789	0.0862	0.2234
<i>canny_thr</i>	100	3514	31.6928	0.0737	0.0985	0.2089
<i>canny_thr</i>	125	3986	43.32	0.0778	0.0948	0.2073
<i>canny_thr</i>	150	4392	51.87	0.0771	0.1062	0.1990
<i>LRE_{min}</i>	30	3514	31.69	0.0737	0.0985	0.2089
<i>LRE_{min}</i>	50	4510	54.89	0.0801	0.1076	0.1944
<i>LRE_{min}</i>	100	5019	63.17	0.0844	0.1119	0.1925
<i>SRE_{max}</i>	0.1	5019	63.17	0.0771	0.1121	0.1926
<i>SRE_{max}</i>	0.25	5019	63.17	0.0766	0.1162	0.1925
<i>SRE_{max}</i>	0.5	5018	63.17	0.0772	0.1222	0.1923
<i>SRE_{max}</i>	0.75	5013	63.00	0.0779	0.1206	0.1922
<i>SRE_{max}</i>	0.9	3514	31.69	0.0737	0.0985	0.2089
<i>SRE_{max}</i>	1	2949	13.95	0.0727	0.0917	0.1987

the specific bounds of the texture descriptors impact the performance of the proposed pipeline.

Specifically, we examine the influence of the *Short Run Emphasis* maximum limit (*SRE_{max}*), the *Long Run Emphasis* minimum threshold (*LRE_{min}*), and the threshold used to obtain the Canny edge map (*canny_thr*). These parameters govern the system's ability to discern between structured surfaces and unstable or homogeneous regions. Concurrently, in each part of the ablation study, the effect of parameter variations on the execution times associated with point extraction and odometry estimation was examined.

This study utilizes a dataset featuring a trajectory that alternates between outdoor and indoor corridors, including laboratory entry and exit. The trajectory starts at an outdoor corridor, followed by an indoor laboratory entry, traversing through an indoor hallway, followed by a return to the initial outdoor location. The sequence is characterized by severe illumination changes at indoor–outdoor transitions, which challenge the robustness of system parameter calibration and make it well suited for the ablation analysis presented in this work. Since ground-truth measurements are unavailable, loop closure is enforced during post-processing by exploiting the known initial and final pose. The resulting trajectory is then used as a reference to quantify pose estimation deviations, expressed in terms of Absolute Trajectory Error (ATE), under different parameter configurations.

Table I summarizes the results of the ablation study. Each configuration is assessed based on the mean number of extracted points per frame, the percentage of frames outside the desired operating range, point retrieval and pose estimation runtimes, and the ATE RMSE with respect to ground truth. The analysis starts from a nominal configuration (*SRE_{max}* = 0.9, *LRE_{min}* = 30, *canny_thr* = 100), and explores the system behavior as individual parameters are varied.

Variations in the *canny_thr* parameter directly affect the number of available points. Lower (more permissive) threshold values allow finer image details to be preserved, resulting in a larger number of extracted points. Under these conditions, the system does not need to aggressively adjust the

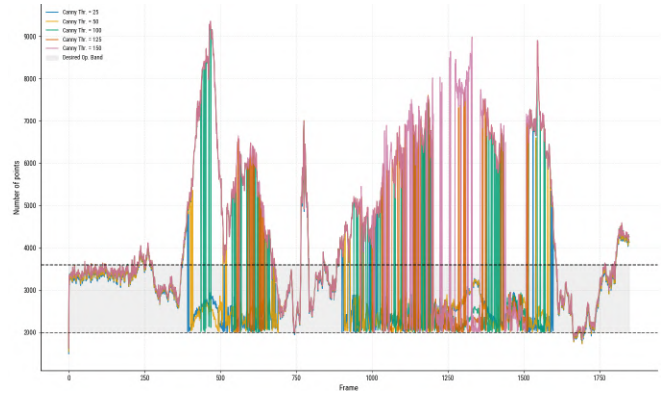


Fig. 9: Evolution of the number of extracted points for different values of the *canny_thr* parameter.

window size to meet the target point count. As shown in Table I, both the average number of extracted points and the percentage of frames outside the operating range increase as the parameter value increases.

This effect is further illustrated in Figure 9. As the Canny threshold increases and edge detail diminishes, the system increasingly narrows its search, often resulting in an excessive number of extracted points over extended intervals.

Additionally, higher values of *LRE_{min}* lead to improved pose estimation performance, suggesting that stricter constraints on the minimum patch response enhance the selection of informative image regions.

Increasing the value of *LRE_{min}* leads to improved pose estimation accuracy, as reflected by the reduction in ATE RMSE. This indicates that enforcing a stricter lower bound on the patch response helps suppress weakly informative regions and promotes the selection of patches that contribute more reliably to direct pose estimation.

In contrast, decreasing *SRE_{max}* to very low values causes most patches to be discarded. For thresholds below 0.75, the system exhibits the highest mean point counts and operating-band violations, as it becomes unable to distinguish between structurally meaningful regions and visually uninformative areas such as distant texture or foliage.

Overall, increasing the number of points supplied to the odometry backend tends to reduce estimation error at the expense of higher computational cost. This trade-off underlines the importance of parameter fine-tuning. The results indicate the absence of a single optimum for all parameters; instead, a balanced configuration yields competitive accuracy while maintaining operational stability and computational efficiency.

C. Comparison with DSO

Direct odometry methods select a subset of pixels through a pixel selection strategy and apply a direct image alignment process to estimate camera motion. One of the most influential contributions on the field was made by Engel et al. in [13], where a Direct Sparse Odometry (DSO) method is proposed to estimate camera motion by minimizing a photometric error function derived from the observed pixel intensities. DSO relies exclusively on image gradient information for pixel selection. Initially, the image is partitioned into 32×32 blocks to compute a gradient threshold given by the median absolute gradient magnitude within each block. This partition is subsequently discarded, and a new partition of the image into $d \times d$ blocks is performed. The pixel exhibiting the maximum gradient magnitude above the previously determined threshold is then selected from each block.

This section presents a comparative evaluation of DSO and the proposed system on several sequences from the TUM RGB-D dataset. Performance is assessed using the Absolute Trajectory Error (ATE), a metric that quantifies global trajectory consistency by measuring the absolute distances between estimated and ground-truth trajectory points [20]. Table II shows the results obtained for the *fr1/plant*, *fr2/desk*, *fr3/structure-texture*, *fr3/structure-no texture*, *fr3/no structure-texture*, and *fr3/no structure-no texture* sequences of the TUM-RGBD dataset.

While sequences such as *fr1/plant* and *fr2/desk* offer rich visual information, they also involve perspective changes that challenge visual odometry algorithms. On the *fr1/plant* sequence, the proposed approach yields results comparable to DSO; in contrast, on the *fr2/desk* sequence, our method lowers the error from 1.7119 to 0.4344, achieving a 74.6% reduction in ATE.

The *structure-texture* family of sequences presents increasingly difficult visual challenges for the pose estimation system. The conditions for visual pose estimation degrade progressively from the most favorable scenario (*fr3/structure-texture*) to the most challenging environment (*fr3/no structure-no texture*).

The *fr3/structure-texture* sequence contains images characterized by abundant structural and textural cues. The proposed approach shows its best performance on this sequence, achieving an ATE RMSE of 0.0296 m, which indicates that, in the presence of rich visual structure and texture, the system can reliably select informative points for direct odometry estimation.

TABLE II: Absolute Trajectory Error (ATE) in meters on TUM-RGBD sequences.

Method	Sequence					
	F1-Plant	F2-Desk	F3-sT	F3-sNt	F3-NsT	F3-NsNt
DSO	0.4840	1.7119	0.8068	0.9605	0.7751	0.3892
Ours	0.5567	0.4344	0.0296	0.1933	0.0574	0.3361

Abbreviations: F1/F2/F3: Freiburg 1/2/3; sT: structure-texture; sNt: structure-no texture; NsT: no structure-texture; NsNt: no structure-no texture.

During the *fr3/structure-no texture* sequence, the camera follows a trajectory around a folding screen covered with white fabric. In this setting, the images contain few visual cues beyond the structural elements of the screen itself. Moreover, in several frames, the distinction between the structure and the background becomes ambiguous in grayscale images. Under these conditions, the proposed system outperforms DSO.

In the *fr3/no structure-texture* sequence, the camera executes a straight trajectory while pointing toward a set of posters placed on the floor, with the structural elements removed. Compared to the previous sequence, both the proposed system and DSO exhibit improved performance due to the presence of distinctive visual features that facilitate point extraction.

Finally, the *fr3/no structure-no texture* sequence presents a highly challenging visual scenario. A similar trajectory is performed; however, the posters are removed, leaving a smooth laboratory floor. In this case, the proposed system achieves performance comparable to DSO, as it is forced to operate in an extreme fallback mode (similar to the case in Figure 5), where points are extracted uniformly across the image. Although suboptimal, this strategy enables the system to maintain tracking at the cost of increased estimation error.

V. CONCLUSIONS

This paper proposes a pixel selection system for pose estimation in direct visual odometry applications. The developed algorithm automatically identifies regions with prominent structural information—features highly desirable for reconstruction and SLAM tasks. By incorporating texture measurements, such as Haralick features, the system dynamically adapts window sizes and filtering thresholds to the navigated environment. This enhances its robustness and adaptability across structured, semi-structured, and unstructured settings.

In highly homogeneous environments where structural features are scarce or absent, a fallback mechanism is activated. This operational mode prioritizes pose-tracking stability over maximum accuracy, thereby preventing catastrophic failures in feature-deprived scenarios.

Comparative results against a standard state-of-the-art method indicate that gradient-based pixel selection should be complemented with additional image characteristics to enhance point extraction on structural elements. Computational timing studies confirm that the proposed processing does not introduce a significant overhead, thereby ensuring its suitability for real-time deployment on UAV platforms.

Although the proposed method demonstrates near real-time behavior under the current experimental setup, strict real-time performance is conditioned on hardware acceleration or implementation in a lower-level, high-performance programming language. Specifically, reimplementing in a compiled language (e.g., C/C++) rather than Python can be the most important improvement to reduce execution overhead and improve computational efficiency. The present Python-based implementation introduces additional computational overhead that may limit execution speed on resource-constrained platforms.

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